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Unlocking the power of sulfur: Essential nutrient to potent bioactives

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Abstract- Sulfur is a vital macronutrient for plants, playing a critical role in their growth, development, and stress response. It's a fundamental building block for vital biomolecules like the amino acid's cysteine and methionine. Plants predominantly absorb sulfate from the soil, often with assistance from beneficial microbes. Beyond its significance in plant physiology, sulfur is crucial across all living organisms, forming integral components of essential amino acids, vitamins such as thiamine and biotin, and other vital biomolecules. This overview focuses on the biological importance of sulfur, particularly highlighting the bioactive organosulfur compounds found in *Allium* species and cruciferous vegetables. Allicin is formed enzymatically from alliin when the plant is mechanically damaged. Allicin is unstable and readily degrades into other organosulfur compounds, including diallyl disulfide (DADS) and diallyl trisulfide (DATS). Similarly, sulforaphane is produced in cruciferous vegetables from glucosinolates through the action of the enzyme myrosinase. These sulfur-containing metabolites exhibit a wide range of potent biological activities. Allicin demonstrates broad-spectrum antimicrobial efficacy against bacteria, fungi, and viruses. Its primary mechanisms involve reacting with thiol groups in microbial proteins and inducing oxidative stress within these pathogens. Allicin and its derivatives show significant anti-cancer properties, actively promoting programmed cell death (apoptosis), inhibiting cell proliferation, and modulating cellular redox states. Their antioxidant capabilities are linked to their interactions with glutathione and their ability to activate the body's natural protective mechanisms. Additionally, these compounds exert anti-inflammatory effects by modulating crucial signaling pathways like Nrf2 and NF-κB. This comprehensive perspective underscores the profound and diverse roles of sulfur and its organic derivatives, highlighting their critical functions in both plant biology and promising therapeutic applications for human health.

Keywords: Nutrient, Allicin, Sulfer, Glutathione, Amino acid

INTRODUCTION

Sulfur is the fourth major nutrient, macronutrient which is essential for the plant growth and development. It plays a crucial role in various physiological processes, from being a structural component of biomolecules to modulating stress tolerance.[1] Sulphur facilitates the performance of plants by improving their growth, photosynthetic, yield and quality parameters.[2] It is a key component of essential amino acids like cysteine and methionine, which are the building blocks of all proteins.[3] Sulfur is necessary for the formation of chlorophyll, the green pigment responsible for photosynthesis, allowing plants to convert sunlight into energy and also helps in cold tolerance in plant due to the process of disulphide linkage.[4] Sulfer is a vital constituent of vitamins such as biotin and thiamine, and plays a role in activating and

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regulating various enzymes critical for plant metabolism. It is part of coenzymes like coenzyme A (CoA), involved in fatty acid metabolism and energy production.[5] It's essential for producing secondary metabolites like glucosinolates, which contribute to plant defense mechanisms against pests and pathogens.[6]

Plants primarily absorb sulfur from the soil through their roots in the anionic form (sulphate). Sulphates SO_4^{2-} are naturally present in soil, often derived from the decomposition of organic matter and mineral sources. Small amounts of sulfur dioxide (SO_2) gas can be absorbed through leaves from the atmosphere, it's generally not a significant source for overall plant nutrition.[7] Some soil microbes, including certain bacteria and fungi (like arbuscular mycorrhizal fungi), can assist plants in acquiring sulfate from the soil, especially under sulfur-depleted conditions.[8]

Biological sources of sulfur

Sulfur is a ubiquitous element found in various forms across the Earth, from geological deposits to living organisms

Living organisms-

Sulfur is an essential element for all living cells. It is a key component of:

Amino acids

Two essential amino acids, methionine and cysteine, contain sulfur and are crucial for protein synthesis. Cysteine stands out as the initial organic compound formed from sulfur.[9-11] It serves as a precursor for synthesizing numerous other sulfur-containing metabolites that are crucial for biological functions, including glutathione (GSH) and methionine. Beyond amino acids, sulfur is found in various essential biomolecules. It's a key component of thiol groups in both proteins and non-protein molecules. Furthermore, sulfur is present in important small biomolecules like iron-sulfur (Fe/S) clusters, the molybdenum cofactor (Moco), and sulfur-modified nucleotides, all of which play significant roles in cellular processes.[12,13]

Vitamins

Some vitamins, like thiamin (B1) and biotin (B7 or H), also contain sulfur. Thiamin is vital for carbohydrate metabolism, converting food into energy, and supporting nerve function. Without sulfur, thiamin wouldn't be able to carry out these critical roles During the biosynthesis of thiamin and biotin, sulfur is transferred from cysteine to cofactor precursors. For thiamin, the sulfur atom of cysteine is transferred to a 66-amino-acid peptide (ThiS) to form a carboxy-terminal thiocarboxylate group. The biotin synthase Fe–S cluster functions as the immediate sulfur donor for biotin formation.[14]

Other organic compounds

Sulfur is an essential mineral for animal health, obtained through their diet. Animals get sulfur from various sources, including plants and animal-based foods. Humans obtain sulfur from a variety of foods, including:

Animal proteins

Meat, poultry, fish, eggs, dairy.[15]

Plant-based foods

Amino acids, methionine, cannot be synthesized by our bodies and therefore has to be supplied by the diet. Cysteine, another sulfur containing amino acid, and a large number of key metabolic intermediates essential for life, are synthesized by us, but the process requires a steady supply of sulfur. Proteins contain between 3 and 6% of sulfur amino acids. A very small percentage of sulfur comes in the form of inorganic sulfates and other forms of organic sulfur present in foods such as *Allium* vegetables (garlic, onions, leeks), cruciferous vegetables (broccoli, cauliflower, cabbage, kale), legumes, nuts, seeds, and whole grains.[16,17]

Biosynthesis of allicin

Allicin, a sulfur-containing natural compound found in Allium species with many different biological properties. This bioactive organosulfur compound responsible for the pungent smell and taste of. It's formed when garlic is crushed or cut, and the enzyme allinase converts allin into allicin.[18] Allicin (allyl 2-propenethiosulfinate or diallyl thiosulfinate or S-allyl cysteine sulfoxide) is a sulfur-containing volatile compound found in white garlic (*Allium sativum* L.) and in other *Allium* species, such as elephant garlic (*A. ampeloprasum* L.), wild garlic (*A. ursinum* L.), field garlic (*A. vineale* L.) and alpine leek (*A. victorialis* L.)[19]

In 1948, Stoll and Seebeck elucidated the thiosulfinate structure of allicin.[19] Natural allicin is formed when the fresh tissue of garlic is enzymatically damaged. The precursor to allicin is alliin (S-allyl-L-cysteine sulfoxide), a non-proteinogenic amino acid. Alliin [(+) -(S)- allyl-L-cysteine-sulfoxide] is enzymatically transformed to allicin in the presence of enzyme alliinase Figure 1. Alliin and alliinase enzyme found in different parts of garlic clove. After crushing the garlic clove, a complex is formed by alliin and enzyme alliinase in the presence of water. The

unstable alliin-alliinase complex is dehydration by pyridoxal phosphate and transformed to allyl sulfenic acid, pyruvic acid and ammonia, as shown in Figure 1.

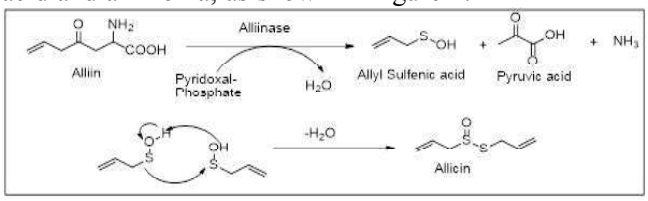


Figure 1- Enzymatic transformation of Alliin to allicin

Allyl sulfenic acid is very reactive at room temperature. Two molecules of allyl sulfenic acid condense togater remove one molecule of water and give allicin. At room temperature, these enzymatic transformations occur in 10-15 minutes.[20] The biosynthetic route to alliin is still not clear. Pioneering work by Granroth[21] reported two possible biosynthetic pathways based on radioactive labelling experiments Figure 2 and Figure 3.

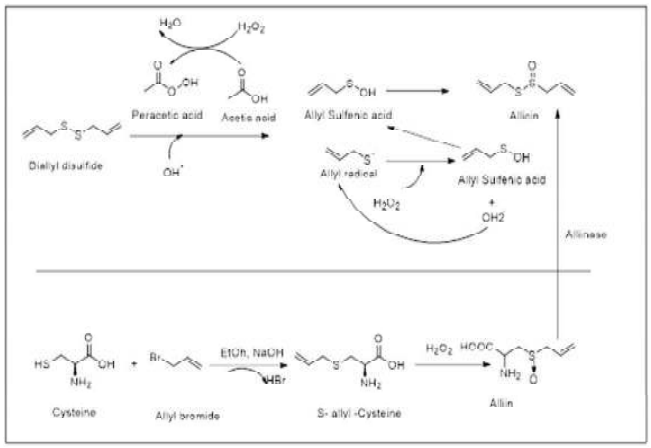


Figure 2- Biosynthesis of allicin

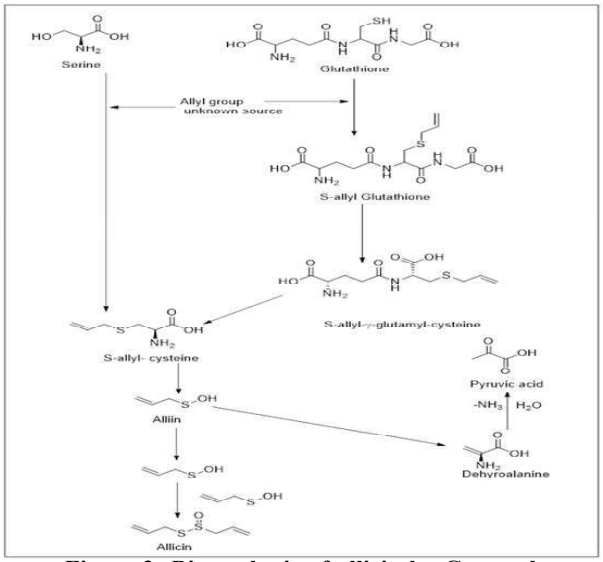


Figure 3- Biosynthesis of allicin by Granroth

Organosulfur compounds – diallyl disulfide, diallyl trisulfide

Garlic and onion and other Allium species produce a group of organosulfur compounds such as diallyl sulfide (DAS), diallyl disulfide (DADS), and diallyl trisulfide (DATS). They are formed from the decomposition of allicin, which is released when garlic cells are broken.[22] These compounds contribute significantly to the pungent aroma of garlic and are responsible for many of its purported health benefits. Allicin itself is quite unstable, particularly in the presence of heat, light, and organic solvents, or even just at room temperature and in aqueous environments. It readily undergoes non-enzymatic decomposition and rearrangement reactions to form a complex mixture of other organosulfur compounds, including DAS, DADS, and DATS.[23,24]

Formation of DAS, DADS, and DATS from allicin

Diallyl disulfide (DADS) is a major organosulfur compound found in garlic oil that is widely used as a flavoring agent.[25] Allicin (a thiosulfinate) contains an unstable disulfide bond with an oxygen atom. Through various rearrangement and decomposition pathways, allicin loses the oxygen and forms the more stable diallyl disulfide. Diallyl trisulfide (DATS) Diallyl sulfide (DAS) compounds.[26-28]

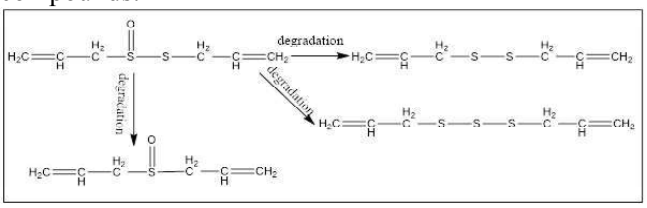


Figure 4- Formation of DAS, DADS, and DATS from allicin

Redox reactions of allicin

Allicin is considered a reactive sulfur species because its unique chemical structure makes it eager to react, especially with other sulfur-containing molecules in our body, particularly those called thiols.[29] Many important proteins and small molecules in our body (like glutathione, a major antioxidant) contain a thiol group, which is a sulfur atom bonded to a hydrogen atom (-SH). Glutathione (GSH) is a small tripeptide found in virtually every cell of your body. It's composed of three amino acids: glutamate, cysteine, and glycine. The crucial part of glutathione for redox reactions is the sulfhydryl group (-SH) of its cysteine residue (Cysteine thiol (-SH). This thiol group can readily donate electrons (get oxidized) and is a powerful nucleophile The highly nucleophilic thiolate anion form

(G-S-) of glutathione attacks the electrophilic sulfur atom of the thiosulfinate group (S(O)) in allicin.[30] This sulfur in allicin is electron-deficient due to the electronegativity of the adjacent oxygen atom. The S-S bond within allicin thiosulfinate group breaks. A new mixed disulfide bond (S-S) is formed between the sulfur of glutathione and one of the allyl-sulfur fragments from allicin Figure 5. This product is called S-allylmercaptoglutathione (GSSA). The sulfur atom of GSH loses electron density as it forms a new disulfide bond. It effectively transitions from a reduced thiol state to a more oxidized disulfide state. The allicin molecule undergoes a net reduction. Its thiosulfinate group is cleaved, and one of its sulfur atoms is reduced to form allyl sulfenic acid (Allyl-SOH)9.[30]

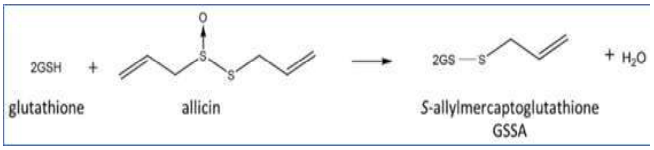


Figure 5- Redox reactions of allicin

Sulforaphane

Sulforaphane is a powerful health-promoting compound found in cruciferous vegetables like broccoli, cauliflower, Brussels sprouts, and kale. These are stored in the plant cells as inactive precursors. It doesn't exist directly as sulforaphane in these plants. These are present in the form of glucosinolates (supfer containing compound). Glucosinolates are phytochemicals that are proposed to be an important contributor to the health benefits of these vegetables.[31,32] Along with glucosinolates, cruciferous vegetables also contain an enzyme called myrosinase (a β-thioglucosidase).[33] Myrosinase and glucosinolates are kept separate in different cellular compartments. Glucosinolates are usually stored in stable condition the vacuole in plant cells. During the preparation, chewing, cutting, chopping, mixing and digestion of foods, glycosylate is released and hydrolysed by enzyme by Myrosinase in to a molecule of β-D-glucose and an unstable aglycone; thiohydroximate-O-sulfonate Figure 6. Thiohydroximate-O-sulfonate is rearranged (Lossen rearrangement) to form different hydrolytic products such as isothiocyanates, nitriles, and other by-products depending on the side chains of glucosinolates and condition of reaction conditions including iron and pH, and myrosinase interacting proteins.[34]

Factors Influencing Sulforaphane Formation:

There are various factors affecting the formation of Sulforaphane these including pre-harvest factors, which can be very important. Although plant species and genotype significantly affect GR content and distribution, the effects of pre-harvest factors, including light, temperature, nutrition (i.e., sulfur (S) and nitrogen (N)), water, and phenological stage.[35-37]

- **Mechanical Damage:** The more the cruciferous vegetable is chopped, chewed, or blended, the more myrosinase is released, leading to greater sulforaphane production.
- **Temperature:** Myrosinase is sensitive to heat. High cooking temperatures (like boiling or microwaving for extended periods) can denature the enzyme, significantly reducing sulforaphane yield. Light steaming for short durations is often recommended to preserve myrosinase activity.
- **pH:** The reaction is favoured at a neutral pH.
- **Presence of Epithiospecifier Protein (ESP):** Some cruciferous vegetables contain an enzyme called Epithiospecifier Protein (ESP). ESP can redirect the hydrolysis of glucosinolates towards the formation of nitriles instead of isothiocyanates (like sulforaphane). However, ESP is more heat-sensitive than myrosinase, so mild heating can inactivate ESP, potentially increasing sulforaphane yield.
- **Gut Microbiota:** Even if the plant's myrosinase is inactivated by cooking, certain bacteria in the human gut possess myrosinase-like activity, allowing for some sulforaphane formation.

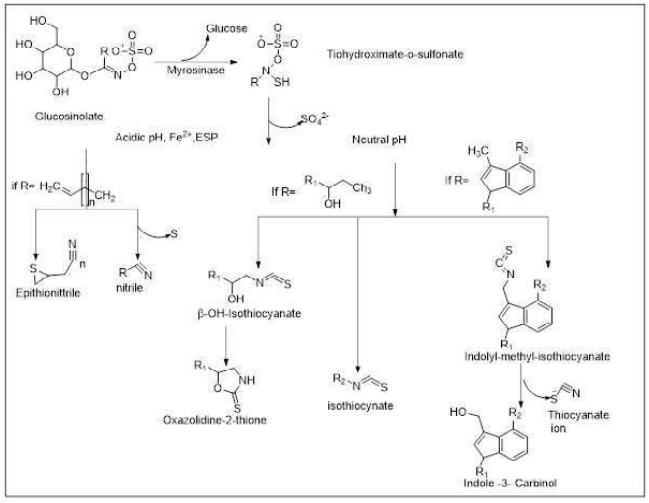


Figure 6- Enzymatic hydrolysis of Glucosinolates and their breakdown products

Antimicrobial activity of allicin

Allicin, a compound found in garlic, exhibits potent antimicrobial activity against a wide range of bacteria, fungi, and viruses. Allicin was found to exhibit antibacterial activity against a wide range of gram-negative and gram-positive, including multidrug-resistant enterotoxico genic strains of *Escherichia coli*, *Enterococcus*, *Shigella dysenteriae*, *S. flexneri*, and *S. sonnei* cells, antifungal activity, particularly against Candida albican*s*, antiparasitic activity, including some major human intestinal protozoan parasites such as *Entamoeba histolytica* and *Giardia lamblia*[39] and antiviral activity against human cytomegalovirus, influenza B, herpes simplex virus type 1, herpes simplex virus type 2, parainfluenza virus type 3, vaccinia virus, vesicular stomatitis virus, and human rhinovirus type.[40]

One of the primary mechanisms through which allicin exerts its effects is its ability to react with thiol groups in proteins and enzymes. Thiols, or sulfhydryl groups (-SH) are often found in the active sites of many enzymes and in the amino acid cysteine within proteins. Allicin can form disulfide bonds with these thiol groups, leading to the modification of enzyme function or protein activity. This can result in the inhibition of microbial enzymes necessary for the survival and replication of bacteria, fungi and viruses.[41-43]

Moreover, allicin is known to induce oxidative stress in microbial cells. The compound can generate reactive oxygen species (ROS) within these cells, disrupting cellular membranes, damaging DNA, and inactivating crucial metabolic enzymes. This oxidative stress is another pathway through which allicin exerts its antimicrobial effects.[44-46]

Anti-cancer activity of allicin

Allicin, the potent organosulfur compound largely responsible for the characteristic odour and many health benefits of freshly crushed garlic, has garnered significant attention for its remarkable anti-cancer activities, Early evidences (1960) demonstrated that allicin could inhibit tumor growth in mice. This was observed when tumor cells, after being cultured with. allicin *in vitro*, were injected into mice, leading to suppressed tumor development.[47]

Allicin can induce cancer cells to form apoptotic bodies and DNA ladders. Both are responsible for programmed cell death (apoptosis).[48] Allicin can alter the balance of oxidation and reduction (redox state) within human cells. This shift leads to the activation of the mitochondrial apoptotic pathway, a major cellular pathway for initiating programmed cell death.[49] Interestingly, allicin's ability to induce apoptosis in colonic cancer cells can be mediated by NF-E2-related factor-2 (Nrf2). This is notable because Nrf2 is usually considered an anti-apoptotic factor that upregulates anti-apoptotic proteins like Bcl-2. This suggests a complex interplay where allicin might exploit or reprogram Nrf2's function in certain cancer contexts to promote cell death.[50,51]

The Precursors and Metabolites od allicin also shows the anticancer properties. Alliin Shows anti-proliferative effects (inhibits cell growth) in human gastric adenocarcinoma cells. It does not negatively affect the growth of normal intestinal cells, indicating a selective action against cancer cells.[52] Diallyl Disulfide (DADS) inhibits human esophageal xenograft tumors (tumors grown in a different species, like mice, from human cancer cells).[53]

Diallyl trisulfide (DATS) blocks the cell cycle (preventing cells from dividing). It inhibits the tumor cell proliferation and angiogenesis.[54] Z-ajoene inhibits the tumor cell growth by: resisting proliferation, stimulating apoptosis and increases the production of peroxide (a reactive oxygen species that can induce cell death.[55] S-allylmercapto cysteine (SAMC) and S-allylcysteine (SAC) plays antitumor roles by inhibiting tumor cell proliferation and inducing apoptosis.[56]

Antioxidation activity of allicin

Allicin has been recognized for its strong antioxidant properties since at least 2006.[57] As a reactive sulfur species, allicin interacts with glutathione (GSH) and protein thiols, depleting intracellular GSH and leading to S-thioallylation of proteins. This reaction is considered central to allicin's biological activity, influencing cellular processes reliant on the reversible oxidation and reduction of protein thiols.[58] Allicin's antioxidant effects are often attributed to two main mechanisms. Firstly, it readily converts to allylsulfenic acid through a Cope elimination reaction, with allylsulfenic acid being a potent antioxidant itself. Secondly, allicin can induce mild oxidative stress, thereby activating the cell's natural protective responses against more severe oxidative damage.[59]

Studies have demonstrated allicin's ability to reduce reactive oxygen species (ROS) production and inhibit various signalling pathways. It also modulates the MAPK pathway to combat oxidation and protects diverse cell

types, including human umbilical vein endothelial cells, oocytes, and neurons, from oxidative stress.[60] Furthermore, allicin shows promise in mitigating oxidative stress and mitochondrial dysfunction in nucleus pulposus cells and has been linked to improved depressive-like behaviours by reducing oxidative stress and enhancing mitochondrial function.[61]

Anti- Inflammatory activity of allicin

Allicin, a key compound in garlic, exhibits significant anti-inflammatory effects through various mechanisms. It mitigates inflammation in conditions like diabetic macroangiopathy by modulating Nrf2 and NF-κB pathways. Allicin also improves osteoarthritis by downregulating PI3K/Akt/NF-κB signaling and alleviates trinitrophenylsulfuric acid-induced inflammation by inhibiting P38, JNK, and NF-κB pathways.[62]

Allicin offers protection against inflammation by suppressing reactive oxygen species (ROS) production and regulating autophagy in fungal infections, and it improves inflammation and oxidative stress in bacterial infections. It also shows hepatoprotective effects against acetaminophen-induced liver injury by reducing oxidative stress and inhibiting inflammatory pathways and hepatocyte apoptosis. Allicin can also enhance immune responses against acute murine malaria. In neurological contexts, it reduces hippocampal neuron apoptosis and depression-like behavior by inhibiting neuroinflammation and the NLRP3 inflammasome.[63]

Allicin, its metabolite Z-ajoene also possesses anti-inflammatory properties, inhibiting pro-inflammatory cytokines (IL-1β, IL-12, IL-6β) and upregulating anti-inflammatory cytokine (IL-10). Z-ajoene achieves this by decreasing STAT3 phosphorylation and nuclear translocation, and by S-thiolating proteins like COX2 and STAT3. The anti-inflammatory actions of garlic's organosulfur compounds, including allicin, are linked to their anti-cancer effects, as they modulate cytokine patterns to inhibit NF-κB, a central regulator linking inflammation to tumor promotion and progression. This shift from a pro-inflammatory to an anti-tumor environment highlights their immune-modulating potential.[64]

CONCLUSION

Sulfur is foundational macronutrient for plants; it helps in physiological processes and growth. Its incorporation into essential biomolecules like amino acids and vitamins highlights its universal biological significance across all living organisms. The focus on bioactive organosulfur compounds, particularly those derived from *Allium* species and cruciferous vegetables, reveals a fascinating intersection of plant biochemistry and human health. These naturally occurring compounds demonstrate a remarkable array of therapeutic properties. Their antimicrobial actions, achieved by targeting microbial thiols and inducing oxidative stress, offer promising avenues for combating various pathogens. Furthermore, their demonstrated anti-cancer, antioxidant, and anti-inflammatory effects, mediated through complex molecular pathways involving redox modulation and signaling pathway regulation, underscore their significant potential in preventative and therapeutic medicine. The comprehensive understanding of sulfur's roles, from its absorption in plants to the intricate bioactivities of its organic derivatives, opens doors for future research.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest relevant to this article**.**

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